INTERNATIONALE
HOCHSCHULE

DLMDSEDE02 Project: Data Engineering
Conception Phase Report

Submitted By

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Project Title: Kisan Call Centre (KCC) Farmer Queries Processing Pipeline

Project Overview

This document outlines the conception phase for a batch-processing data system designed to handle farmer query data from the Kisan Call Centre (KCC) (*Kisan means Farmer*) made available by the Open Data Portal from the Indian Government.

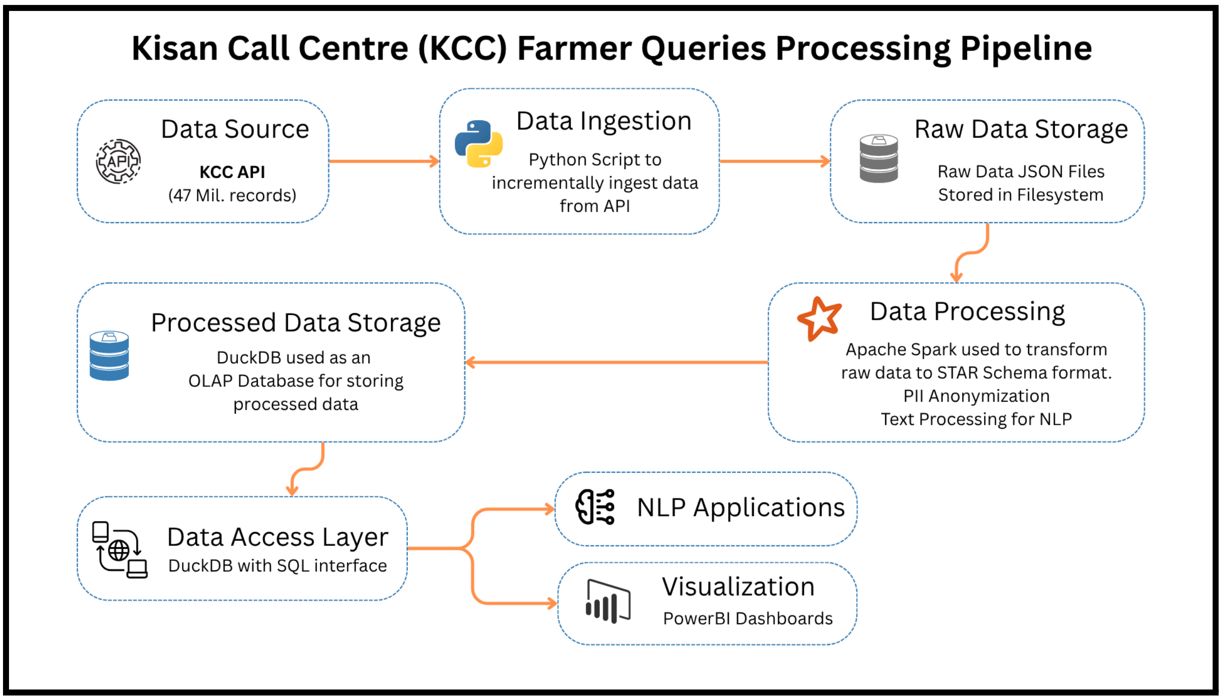
The system will ingest, process, and analyze approximately **47 million records** of farmer queries to provide valuable insights for agricultural planning and support, with specific focus on enabling NLP applications.

Expected Outcomes

Enables Evidence-Based Decision Making
By processing and analyzing 47 million farmer queries, the system provides actionable insights into the real challenges and information needs faced by Indian farmers. This supports government agencies, agricultural planners, and policymakers in making informed decisions.

Enables Natural Language Processing (NLP)
The pipeline prepares and cleans text data, making it ready for NLP applications like sentiment analysis, topic modeling, and automated query classification.

Architecture Diagram



Components of the Architecture

1. Data Source

The KCC API provides the raw data, consisting of approximately 47 million records of farmer queries collected by the Kisan Call Centre. The data is organized by district and month and includes both the questions asked by farmers and the responses provided by agricultural experts.

Data Source Link:
https://www.data.gov.in/catalog/district-wise-and-month-wise-queries-farmers-kisan-call-centre-kcc

2. Data Ingestion

Python scripts (using the requests library) are used to incrementally fetch new data from the KCC API. The scripts are scheduled (e.g., via cron jobs) to regularly pull data for each year and month, ensuring efficient extraction and robust error handling.

3. Raw Data Storage

The ingested data is stored as raw JSON files on the local filesystem, organized by year and month. This storage preserves the original API responses in an immutable format for traceability and reprocessing if needed.

4. Data Processing

Apache Spark (PySpark) processes the raw data in monthly batches, partitioned by year, month, and state. The processing includes:

- Data cleaning
- Anonymization of personally identifiable information (PII)
- Creation of a star schema (fact and dimension tables)
- Text preprocessing to prepare for NLP applications

5. Processed Data Storage

The cleaned and transformed data is loaded into DuckDB, an OLAP (Online Analytical Processing) database. DuckDB is optimized for analytical workloads, supports efficient columnar storage, and is used to store the star schema for downstream analysis.

6. Data Access Layer

DuckDB provides a SQL interface for easy querying of the processed data. This layer serves as the central access point for data analysts, visualization tools, and NLP applications, ensuring secure and controlled access.

7. NLP Applications

Processed and pre-cleaned text data is made available for Natural Language Processing (NLP) applications. These applications can extract insights, perform sentiment analysis, or classify queries to support agricultural planning and policy-making.

8. Visualization

Power BI dashboards are built on top of the processed data to provide interactive visualizations. These dashboards offer insights such as geographical distribution of queries, seasonal trends, and query categorization, supporting decision-making for stakeholders.

Additional Details

Suitability of Star Schema

I have chosen the star schema model for this data for the reasons below

1. **Query Pattern Match**
    - Star schema excels at answering multi-dimensional analytical questions (e.g., "What are the most common queries about rice crops during monsoon season in Maharashtra?")
    - Supports aggregation queries that will be common in agricultural analysis
2. **Dimensional Nature of Data**
    - The KCC data has clear dimensional attributes (location, time, crop, category, sector)
    - These dimensions have natural hierarchies (State > District > Block, Year > Month > Day)
3. **Performance Benefits**
   Simplified join paths improve query performance for analytical workloads
    - DuckDB's columnar storage works efficiently with star schema designs
    - Dimensional filters can leverage partitioning strategy (year > month > state)

Reliability, Scalability, and Maintainability Aspects

Reliability

- Implement retry logic, error handling, and data validation in ingestion scripts.
- Use immutable raw storage for traceability.

Scalability

- Use Apache Spark for distributed processing and partition data by year/month/state.
- Containerize each microservice with Docker for horizontal scaling.

Maintainability

- Modularize codebase, use CI/CD pipelines for automated deployment, and maintain clear logging and monitoring for all services.

Data Security, Governance, and Protection Measures

Data Security

- Anonymize PII during processing
- Use access control on DuckDB, and encrypt data at rest and in transit.

Governance

- Maintain audit logs for data access and processing
- Enforce schema validation
- Document data lineage.

Protection

- Regularly back up both raw and processed data
- Restrict access to sensitive data.